F13-SW-103
2000-F13-SLD-EL-10911

F13-SR-001
415V SAG MILL SWITCHROOM

SWR EARTH BAR

FEEDER FROM F13-SW-002
2000-F13-SLD-EL-10301

TO F13-SW-002
2000-F13-SLD-EL-10301

F13-SW-102
SAG MILL 415V ESSENTIAL SWITCHBOARD

EARTH BAR

NEUTRAL BAR
NOTE 1

AFD

2A
63A

2A
63A

MANUAL
MCCB CHANGEOVER
SWITCH 3 POLE 630A

00F-05

415V 3NE~ 50Hz 630A 65kA for 1sec

INC

01F-16	01F-05	01F-21	01F-12	01F-09	02F-16	02F-11	02F-06			
160A	32A	250A	100A	63A	250A	250A	250A			
2A 32A	2A 32A	2A 32A	2A 32A	2A 32A	2A 32A	2A 32A	2A 32A			
/ >	/ >	/ >	/ >	/ >	/ >	/ >	/ >			
90A	70A	90A	70A	70A	90A	90A	90A			
160A 250A	32A 32A					160A 250A	160A 250A			
			ECR							

69kW		10kW	10kW	69kW	CR 48kW	CR 15.9kW	69kW	69kW	69kW	
F13-DB-001	ISL	F13-DB-201	F13-UPS-001	F13-DB-304	F13-CR-402	F13-CR-411	F13-DB-307	F13-DB-308	F13-DB-003	
SAG MILL 415V SWITCHROOM L&SP DB	F13UPS001-ISL 240VAC UPS MAINTENANCE BYPASS SWITCH	SAG MILL 415V SWITCHROOM 240VAC UPS DB	SAG MILL 415V SWITCHROOM 240VAC UPS	MILLING AREA 1 L&SP DB	CLASSIFICATION AREA OVERHEAD CRANE	MILL LINER MAINTENANCE HOIST 1	MILLING AREA 4 L&SP DB	MILLING AREA 5 L&SP DB	SAG & BALL MILL VSD SWITCHROOM L&SP DB	
2000-F13-SCM-EL-10305	2000-F13-SCM-EL-10306	2000-F13-TLD-EL-10012	2000-F13-TLD-EL-10012	2000-F13-SCM-EL-10307	2000-F13-SCM-EL-10308	2000-F13-SCM-EL-10309	2000-F13-SCM-EL-10364	2000-F13-SCM-EL-10973	2000-F13-SCM-EL-10906	

NOTES

1. EARTH-NEUTRAL LINK INSTALLED IN UPSTREAM F13-SW-002 INCOMER TIER. SIGNAGE SHALL BE PLACED ON F13-SW-102 SWITCHBOARD READING "M.E.N. LINK LOCATED IN F13-SW-002 INCOMER TIER"
2. SWITCHBOARD SHALL BE EXTENSIBLE AND BE PROVIDED WITH THE FACILITY TO EXTEND THE MAIN BUSBAR AND APPEND ADDITIONAL TIERS AT ONE END WITHOUT MAJOR STRUCTURAL DISASSEMBLY.
3. MINIATURE CIRCUIT BREAKERS SHALL BE SIZED AND RATED TO SUIT ACB AND INCOMER REQUIREMENTS.
4. CIRCUIT BREAKER TO BE SUBSTITUED FOR FUSED-SWITCH WHERE REQUIRED BY EQUIPMENT MANUFACTURERS GUIDELINES.

LEGEND

PFR	PHASE FAILURE RELAY
/ >	EARTH LEAKAGE RELAY
xxxA / xxxA	ST TRIP SET POINT / CB SIZE
AFD	ARC FLASH DETECTION
ECR	EARTH CONTINUITY RELAY

NORTHERN STAR RESOURCES LIMITED

Approved	X
Approved as Noted	
Not Approved	
For Information Only	

Signed [redacted] Date 15/09/2025

ACCEPTANCE DOES NOT RELIEVE THE SUPPLIER / CONTRACTOR OF ITS RESPONSIBILITY OR LIABILITIES UNDER THE PURCHASE ORDER / CONTRACT

DRG No.	REFERENCES	REV	DATE	DESCRIPTION	DRAW	CHECK	DESIGN	TECH APP	PROJ APP
2000-F13-SCM-EL-10304	F13-SW-002 415V MCC ARC FLASH SYSTEM SCHEMATIC DIAGRAM								
2000-F13-SCM-EL-10303	F13-SW-002 415V MCC 24VDC DISTRIBUTION SCHEMATIC DIAGRAM								
2000-F13-SCM-EL-10354	F13-SW-102 415V ESSENTIAL SWITCHBOARD INCOMER SCHEMATIC DIAGRAM								
2000-F13-DRG-EL-10101	F13-SR-001 415V SWITCHROOM ELECTRICAL EQUIPMENT LAYOUT								
2000-F13-BLK-IC-10001	F13-SR-001 415V SWITCHROOM AND REMOTE I/O COMMUNICATIONS BLOCK DIAGRAM								
2000-F13-SLD-EL-10301	F13-SW-002 415V MCC SINGLE LINE DIAGRAM SHEET 1								
2000-F00-SLD-EL-10002	F00-SW-010 33kV SWITCHBOARD SINGLE LINE DIAGRAM SHEET 1 OF 5	1	18AUG25	FAT REDLINES INCORPORATED				MK	
2000-F00-STD-EL-10001	GENERAL NOTES AND STANDARD DRAWINGS INDEX	0	29AUG24	APPROVED FOR CONSTRUCTION					

NORTHERN STAR RESOURCES LIMITED

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PRIMERO PROJECT NUMBER: 23517

PROJECT: NORTHERN STAR RESOURCES LTD
KCGM GROWTH PROJECT

TITLE: MILLING & CLASSIFICATION - SAG MILL
F13-SW-102 415V ESSENTIAL SWITCHBOARD
SINGLE LINE DIAGRAM

SHEET	SCALE	DRG No.	REVISION
A1	NONE	2000-F13-SLD-EL-10311	1

O:\CAD PROJECTS\235 NORTHERN STAR RESOURCES LTD\23517 KCGM GROWTH PROJECT\03 DELIVERABLES\E\LV\F13 - MILLING & CLASSIFICATION\2000-F13-SLD-EL-10311.DWG 27AUG25 7:52AM HOTSEAT04